

in the website reflects the primary keywords, and it will form the basis of how the website is indexed and perceived by search engines for years to come. The better the keywords, the greater the role these will play in the content and marketing strategies to gain a high search engine ranking.

2. **Anchor text phrase:** The anchor text or link label appears as the clickable, underlined part of a link from search engine results or on a friend's website. The choice is crucial due to the availability of few words. The link label should make sense and sound good for optimisation purposes. On your own website, if you go to *Page Options > Search Engine Optimization > Page Title* you can place this phrase so that each page in your website will be seen by Google and be clickable as the 'blue text'.
3. **Backlinks and the link page:** It is the link from another website with proper anchor text that links the page to a page on your main website. Backlinks look at the quality and reputation of websites. It requires significant research to get the best backlink — searching for competitors and then investigating which backlink domains are associated so that the website is connected to those same domains.
4. **Competition placement:** After researching keywords and anchor text, it is important to track the rank of the website in comparison with other websites in the same industry. You must put keywords in the search engine between “ ” and note down the competing pages, before using search triggers to optimise the website.
5. **Analytics and tracking tools:** Analytics and tracking tools provide insights on how a site is performing. Analytics measure traffic, page views, clicks, and so much more. These resources collect and analyse your data, as well as your competitors' data. With this information, you can better understand your customers and apply that knowledge to improve your site for a greater online experience and increased conversions. Different analytics produce different data, but the underlying goal is the same — to measure the performance of your Web presence.
6. **Layout of the page and calls to action:** Page layout plays a crucial role in how the user navigates to various sub-pages, and indexing makes it easy to crawl through the information. A call to action (CTA) is a prompt on a website that tells the user to take some specified action. A CTA is typically written as a command, such as 'Sign Up' or 'Buy Now' and generally takes the form of a button or hyperlink. Both are good for Web optimisation.
7. **Design and branding 'your look':** This is where an optimisation strategy should start to leverage website searches, and extend the website with a professional design. It is recommended that you choose the best template with regard to style and layout, leaving the content intact.

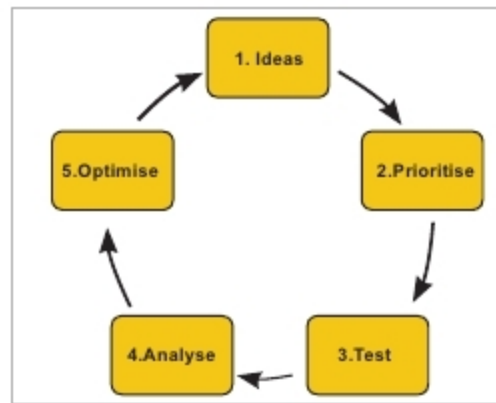


Figure 1: Web optimisation process

Website design should be optimised to be viewed on laptops, PCs, smartphones, tablets and even smart TVs.

8. **Database/forms:** The database and forms should be customised as per the end user's needs. It is recommended that you have a proper backup of the database, and it should be optimised to support all types of queries, emails, newsletters, etc.

9. **Auto-response systems:** Eighty per cent of your communications with prospective

customers or audience can be semi-automated by using forms, databases and auto-responder software, or third-party mailing and re-mailing systems. Thinking through exactly what you are offering to your website's visitors is key to the success of a website optimisation strategy. This should be done once you have performed sufficient analysis, and have built trust with the visitors to your website with your brand and layout.

10. **E-commerce:** This is the last stage of Web optimisation and the website should support all e-commerce systems. Website optimisation for e-commerce ought to be achieved after a thorough study of your website's visitors and their purchasing habits.

To perform Web optimisation is not an easy task, as it involves a number of varied procedures and steps. A lot of commercial and open source tools are available for Web optimisation, but in this article, only the top open source tools for Web optimisation are highlighted to assist Web developers.

Locust.io

Locust is an open source, easy to use, distributed Web optimisation tool for load testing of the website. It is designed to determine how many concurrent users a particular system can handle. It works on the same pattern as a swarm of locusts attacking the website, and the behaviour of every locust represents a user and the swarming process is monitored from the Web UI in real-time. Locust.io is completely event based and, therefore, it's possible to support thousands of concurrent users on a single machine. In contrast to many other event based apps, it doesn't use callbacks. Instead, it uses lightweight processes, through *gevent*.

Features:

- Highly distributed and scalable, and supports thousands of users because of its event based nature.
- Neat HTML+JS UI to give entire information in real-time.
- Facilitates writing test case scenarios in Python code.
- Being Web-oriented, it can test any type of system.
- It is a lightweight, small and easily hackable tool facilitating Web developers to give the best Web loading results.

Official website: <https://locust.io/>

Latest version: 0.9.0